

ECE 657D: Example Project Ideas

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Already have a project idea?

A popular request at the beginning of the term was for guidance on project ideas—this document is aimed at those students. If you already have an idea for a project you’re excited about, you are welcome to close this PDF and get started on your own idea right away.

The following are example project ideas to help you get started. You are **not limited** to these topics—they are meant to illustrate the expected scope and style. You are encouraged to pursue a question that interests you, whether or not it appears on this list.

Each idea includes a brief description and a suggested starting reference. A good project will go beyond simply reproducing a paper; it should involve some original element, such as a new experiment, a comparison across methods, an application to a new dataset, or an ablation study.

Vision

1. Data Augmentation Strategies for Small Datasets

Compare data augmentation techniques (e.g., random cropping, CutOut, MixUp, CutMix) on a small image classification dataset. Measure how each strategy affects generalization as a function of training set size.

Starting reference: Zhang et al., “mixup: Beyond Empirical Risk Minimization” (ICLR 2018)

2. Lightweight Image Classification with Knowledge Distillation

Train a small “student” network to mimic the predictions of a larger “teacher” network. Compare the student’s accuracy and efficiency to the teacher and to a student trained from scratch.

Starting reference: Hinton et al., “Distilling the Knowledge in a Neural Network” (NeurIPS 2014 Workshop)

3. Neural Style Transfer: Content–Style Trade-offs

Implement neural style transfer and systematically study how the content/style loss weighting affects output quality. Propose a simple method for automatically balancing the two losses.

Starting reference: Gatys et al., “A Neural Algorithm of Artistic Style” (2015)

Natural Language Processing

4. Prompt Engineering for Text Classification

Compare zero-shot and few-shot prompting strategies for a pre-trained language model (e.g., GPT-2 or a small LLaMA variant) on a standard text classification benchmark. Analyze how prompt format and example selection affect accuracy.

Starting reference: Brown et al., “Language Models are Few-Shot Learners” (NeurIPS 2020)

5. Attention Head Pruning in Transformers

Investigate which attention heads in a pre-trained transformer can be removed with minimal performance loss. Implement a simple pruning strategy and evaluate on a downstream task.

Starting reference: Michel et al., “Are Sixteen Heads Really Better than One?” (NeurIPS 2019)

6. Retrieval-Augmented Generation for Question Answering

Build a simple RAG system that retrieves relevant passages from a knowledge base and conditions a language model on them to answer questions. Compare against a closed-book baseline.

Starting reference: Lewis et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks” (NeurIPS 2020)

Generative Models and Theory

7. VAE Latent Space Geometry

Train VAEs with different values of β (the KL weight) on MNIST or CelebA. Visualize and quantitatively analyze how the latent space geometry changes: measure disentanglement, interpolation quality, and reconstruction-generation trade-offs.

Starting reference: Higgins et al., “beta-VAE: Learning Basic Visual Concepts with a Constrained Variational Framework” (ICLR 2017)

8. Denoising Diffusion on a 2D Toy Dataset

Implement a simple DDPM on 2D synthetic data (e.g., spirals, moons, mixture of Gaussians). Visualize the forward and reverse diffusion processes. Study how the number of diffusion steps and the noise schedule affect sample quality.

Starting reference: Ho et al., “Denoising Diffusion Probabilistic Models” (NeurIPS 2020)

9. Neural Scaling Laws on Small Models

Train a family of small language models (varying parameter count, dataset size, and compute budget) and measure how loss scales with each factor. Compare your empirical scaling exponents to those reported in the literature.

Starting reference: Kaplan et al., “Scaling Laws for Neural Language Models” (2020)

Applications

10. Medical Image Segmentation with U-Net

Train a U-Net on a publicly available medical imaging dataset (e.g., retinal vessel segmentation, lung CT segmentation). Investigate the effect of data augmentation and pre-training on segmentation quality.

Starting reference: Ronneberger et al., “U-Net: Convolutional Networks for Biomedical Image Segmentation” (MICCAI 2015)

11. Graph Neural Networks for Molecular Property Prediction

Apply a graph neural network (e.g., GCN, GAT, or MPNN) to predict molecular properties from the ZINC or QM9 dataset. Compare graph-based representations to fingerprint-based baselines.

Starting reference: Gilmer et al., “Neural Message Passing for Quantum Chemistry” (ICML 2017)

12. Time Series Forecasting with Transformers

Compare a transformer-based model against LSTMs and classical baselines (ARIMA) on a time series forecasting benchmark (e.g., weather, energy, or traffic data). Analyze where attention-based models help and where they struggle.

Starting reference: Zhou et al., “Informer: Beyond Efficient Transformer for Long Sequence Time-Series Forecasting” (AAAI 2021)

Ethics and Safety

13. Measuring and Mitigating Bias in Image Classifiers

Train a face attribute classifier on CelebA and measure performance disparities across demographic groups. Implement a simple bias mitigation technique (e.g., re-weighting, adversarial debiasing) and evaluate its effectiveness.

Starting reference: Madras et al., “Learning Adversarially Fair and Transferable Representations” (ICML 2018)

14. Differential Privacy in Deep Learning

Train a neural network with differentially private SGD (DP-SGD) and measure the privacy–utility trade-off. How much accuracy is lost at various privacy budgets ϵ ?

Starting reference: Abadi et al., “Deep Learning with Differential Privacy” (CCS 2016)

15. Adversarial Robustness of Image Classifiers

Implement FGSM and PGD adversarial attacks on a trained image classifier. Evaluate adversarial training as a defense and measure the trade-off between clean accuracy and robust accuracy.

Starting reference: Madry et al., “Towards Deep Learning Models Resistant to Adversarial Examples” (ICLR 2018)

Note: These are starting points. A strong project will narrow the scope to a specific, well-defined question and go deeper rather than broader. Feel free to visit office hours if you would like to discuss your project idea.